Seminar4

luni, 1 noiembrie 2021 12:07

Proprietăți de închidere pe LR

op = operație pe limbaje

$L_1 \ op \ L_2 = L_3$ / $L_1, L_2 \in LR$ / $\Rightarrow L_3 \in LR$

$\cup, \cap, \circ, *, \setminus, C$

$L_1, L_2 \in LR$ at. $L_1 \cup L_2 = L_3 \in LR$

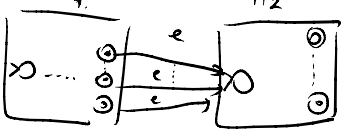
$L_1 \cap L_2 \in LR$

$L_1 \circ L_2 \in LR$

$L_1^*, L_2^* \in LR$

$L_1 \setminus L_2, L_2 \setminus L_1 \in LR$

$CL_1, CL_2 \in LR$

Operație	ER: $E_1 \to L_1$, $E_2 \to L_2$ $L_1, L_2 \in LR \Rightarrow$	AFD $L_1, L_2 \in CR \Rightarrow L_1 \to M_1$, $L_2 \to M_2$ $M_i = (K_i, \Sigma_i, \delta_i, s_i, F_i)$	AFN $L_1 \to M_1$, $L_2 \to M_2$ $M_i = (K_i, \Sigma_i, \Delta_i, s_i, F_i)$
∘	$L_3 = L_1 \circ L_2$ $E_3 = E_1 \cdot E_2$ $\mathcal{L}(E_3) = L_3$	— AFD $L_1 \circ L_2$.	 $M_3 = (K_1 \cup K_2, \Sigma_1 \cup \Sigma_2, \Delta_1 \cup \Delta_2 \cup \{(f_1, e, s_2) / f_1 \in F_1\}, s_1, F_2)$

AFD pt. $L_1 \circ L_2$:

1. AFN pt. L_1
2. AFN pt. L_2
3. concatenez M_1 cu M_2
4. Transform AFN M_3 în AFD

3. Concatenare a două limbaje

4. Transformă AFN M_3 în AFD

$\Sigma = \{a, b\}$

$L_1 = \{ w_1 \in \Sigma^* / w_1 \text{ conține } a \}$ $\quad \Sigma_1 = \{a\}$

$L_2 = \{ w_1 \in \Sigma^* / w_1 \text{ conține } b \}$ $\quad \Sigma_2 = \{b\}$

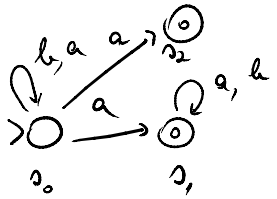
L_1 : $E_1 = (a \cup b)^* a (a \cup b)^*$

L_2 : $E_2 = (a \cup b)^* b (a \cup b)^*$

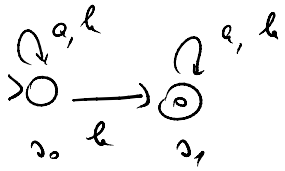
L_3 : $E_3 = E_1 \circ E_2 = (a \cup b)^* a (a \cup b)^* (a \cup b)^* b (a \cup b)^*$

$L_3 = \{ w_1 \in \Sigma^* / w_1 = w_{1_1} w_{1_2}, w_{1_1} \text{ conține } a \}$
w_{1_2} conține b

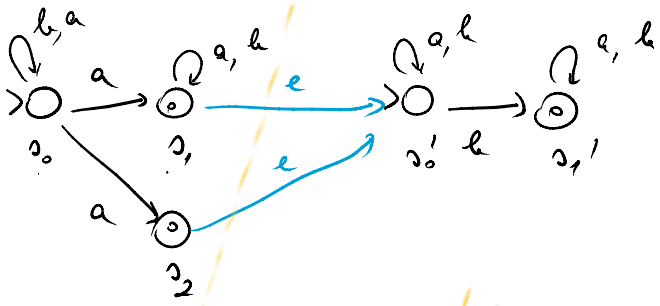
M_1 :



M_2 :



$L_3 = L_1 \circ L_2$: M_3 :



$K_3 = K_1 \cup K_2$

L_2

L_1

M_1

M_2

*	$L_1 \to E_1$ $L_3 = L_1^*$	—	M_1	M_3

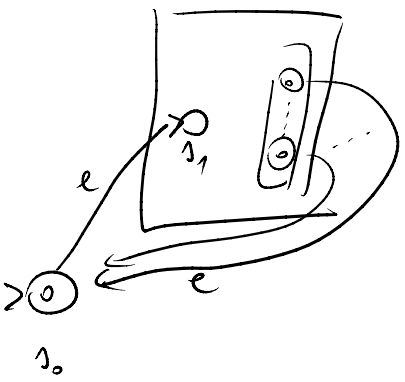
$L_3 = L_1^*$

$E_3 = E_1^*$

$E_1 = ab$

$E_3 = (ab)^*$

M_1 M_3



$M_3 = (K_1 \cup \{s_0\}, \Sigma_1, \Delta_1 \cup \{(s_0, e, s_1)\} \cup \{(f_1, e, s_0) / f_1 \in F_1\}, s_0, F_1 \cup \{s_0\})$

$\cup$	✓	✓	✓
$\cap$	—	✓	—
$\setminus$	—	✓	—
C	—	✓	—

LR $\cup, \cap, \setminus, C, \circ, ^*$

1. Fie $L_1 = \{a^n b^n / n < 5\}$

$L_2 = \{a^n b^m / n \geq 0, m \geq 0\}$

$L_3 = \{a^n b^m c^p / n = m, n \geq 0, p \geq 0\} \notin LR$

$\cap$, limbaiele $L_1, L_2, L_1^*, L_2^*, L_3^*$

Cum sunt limbajele L_1, L_2, L_3, L_1^*, L_2^*, L_3^*,

$L_1 \cup L_2$, $L_1 \cap L_3$, $L_1 \cdot L_3$,

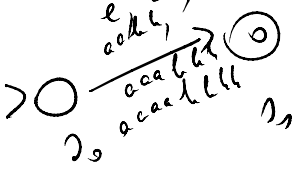
$L_3 \setminus L_2$? . $L_1 \cap L_3 = L_1 \in LR$.

$L_1 = \{ e, ab, aabb, aaabbb, aaaabbbb \}$

$E_1 = e \cup ab \cup aabb \cup aaabbb \cup aaaabbbb$

$\forall L$, dacă $|L| < \infty$ at. $L \in LR$.

$\Rightarrow L_1 \in LR$.

M_1 : 

$L_2 \rightarrow E_2 = a^* b^*$ $\Rightarrow L_2 \in LR$.

$L_1 \in LR$

$L_2 \in LR$

$L_3 \notin LR$.

O teoremă LR o respectă

$L_1 \in LR$

+ este prop. de închidere pe LR $\Rightarrow L_1^* \in LR.$

$L_2^* \in LR.$

$L_1 \in LR$

$L_2 \notin LR$ $\quad L_1 \cup L_2 \notin LR.$ fals.

L_1

$L_2 = \{ a^n b^m \quad n \geq 0, \quad m \geq 0 \}$

$L_3 = \{ a^n b^m c^p \mid n = m, \quad m \geq 0, \quad p \geq 0 \}$

$L_2 \cup L_3 = \{ a^n b^m c^p \mid n = m, \quad p \geq 1 \} \cup$

$\{ a^n b^m \} \notin LR.$

$L_2 = \{ a^n b^m, \quad n, m \geq 0 \} \in LR \quad a^* b^*$

$L_2 = \{ a^n b^{\cdots}, \; n, m \geq 0 \} \in LR$ a ↻

$L_3 = \{ a^n b^n, \; n \geq 0 \}$

$L_2 \cup L_3 = L_2 \in LR.$

$$L_3 \setminus L_2 = \{ a^n b^m c^p \,/\, n = m, \; p \geq 1 \}$$

2. Fie $L_1 \cup L_2 \in LR$ și $|L_1| < \infty$

L_1 e finit.

Ce s poate spune despre L_1 și L_2?

$L_1 \in LR.$

L_2 poate fi orice

$L_1 \cup L_2 = L_3 \in LR$

$L_1 \in LR$ și finit

$\overline{L_2} = \underbrace{(L_3 \setminus L_1)}_{LR.} \cup (L_1 \cap L_2)$ / $\Rightarrow LR.$

$L_1 \cap L_2$ finit $\in LR.$

$L_1 = \{ a, b \}$ $\quad L_3 = \{ a, b, c \}$

$L_2 = \{ b, c \}$ $\quad L_3 \setminus L_1 \neq L_2$

$L_1 = \{ \varepsilon \}$

$L_2 = a^+$

Fi multimea

$$N = \{ L \subset \Sigma^* \mid \forall w \in L, \#_a(w) > 0 \}$$

În raport cu care operații este multimea N închisă?

$\cap, \cup, \cdot, *, \overline{x}$

$L_1, L_2 \in N$ $\quad L_1 \cap L_2 \in N$

$L_1 \cup L_2 \in N$

$L_1 \in N$

$L_1 \cup L_1 = L_1 \in N$

$L_1, L_2 \in N$ $\quad L_1 \cup L_2 = L_3 \in N$? ✓

$L_1 \cap L_2 = L_3 \in N$?

$L_1 \circ L_2 = L_3 \in ?$

L_1^* $\times$

$L_1 \times L_2$ (ab, ba)

$A_1 = \{1, 2\}$

$A_2 = \{3, 4\}$

$A_1 \times A_2 = \{(1,3), (1,4), (2,3), (2,4)\}$